



Wenhao Wu (吴文浩)

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Research Interests

Large Language Model Agents, Retrieval-Augmented Generation, Meta Reinforcement Learning, and Embodied AI.

Education

Nanjing University, School of Management and Engineering *09/2024–Present*
M.Eng. in Control Engineering; supervised by Prof. Zhi Wang.

Nanjing University, School of Management and Engineering *09/2020–06/2024*
B.Eng. in Automation.

Research Experience

- **MemFuse: Multi-Source Memory Fusion from Fragmented Observations**
arXiv preprint, 2026; co-author
Chao Li, Yuanfa Li, **Wenhao Wu**, Xule Liu, Zhi Wang, Kun Shao. Introduces MemFuseBench for multi-source memory fusion and a dual-layer causal fusion graph that preserves source provenance while organizing fragmented observations into retrievable memories.
- **From Conflict to Consensus: Boosting Medical Reasoning via Multi-Round Agentic RAG**
ICML 2026; first author
Wenhao Wu, Zhentao Tang, Yafu Li, Shixiong Kai, Mingxuan Yuan, Zhenhong Sun, Chunlin Chen, Zhi Wang. Proposed conflict-guided multi-round retrieval for medical reasoning, using disagreement among sampled responses to generate precise queries and sequence-level entropy or a lightweight BERT verifier to rerank historical reasoning trajectories.
- **HarnessPAI: An Evolving Harness for Physical AI**
arXiv preprint, 2026; co-author
Xin Wang*, **Wenhao Wu***, Menghao Zhang*, Zhi Wang, Kun Shao, Jian Luan, Yang Li, Qing Li, Shangding Gu, Huichi Zhou, Shuqing Shi, Fei Ni, Shuo Lu, Weicheng Meng, Kang Li, Jin Wu, Kang Zhao, Shangmin Guo, Gen Li, Yongqiang Tang, Zhizhong Zhang, Yuan Xie, Heng Qu. Introduced a model- and embodiment-agnostic harness for Physical AI in which the harness and action model co-evolve through rollout feedback: action primitives are organized into executable programs, failures are distilled into reusable skills, and the system improves without retraining the underlying action model.
- **Look Before You Leap: Distilling Tree Search into Action Evaluation for Frozen VLA Models**
arXiv preprint, 2026; co-author
Xinyi Xie, Zican Hu, Zhanyu Liu, Yicheng Dong, **Wenhao Wu**, Haoran Li, Chunlin Chen, Zhi Wang*, Pichao Wang. Proposed SVA (Search, Value, and Act), which keeps the VLA backbone frozen and distills Monte-Carlo tree search into a lightweight Q-value evaluator that selects the best of N candidate actions at deployment without simulator access.
- **Mixture-of-Experts Meets In-Context Reinforcement Learning**
NeurIPS 2025; first author
Wenhao Wu, Fuhong Liu, Haoru Li, Zican Hu, Daoyi Dong, Chunlin Chen, Zhi Wang. Proposed T2MIR, a dual-MoE framework with Token-wise and Task-wise experts for multi-modal trajectory processing and multi-task in-context RL.
- **Text-to-Decision Agent: Offline Meta-Reinforcement Learning from Natural Language Supervision**
NeurIPS 2025; co-first author
Shilin Zhang*, Zican Hu*, **Wenhao Wu***, et al. Aligned trajectory decision embeddings with natural-language supervision through a world-model encoder, contrastive LoRA adaptation, and conditional policies.

- **Meta-DT: Offline Meta-RL as Conditional Sequence Modeling with World Model Disentanglement**

NeurIPS 2024; student second author

Zhi Wang, Li Zhang, **Wenhao Wu**, Yuanheng Zhu, Dongbin Zhao, Chunlin Chen. Introduced a context-aware world model and self-guided/complementary prompts to improve Decision Transformer generalization to new tasks.

Competition Experience

TRACE: A Self-Evolving Skill Bank for Consistent, Limit-Aware LLM Agents

Technical Report | Homepage | Code | Leaderboard

Darwin Agent · **Wenhao Wu**^{*}, Menghao Zhang^{*}, Xin Wang^{*}, Zhi Wang, Kun Shao, Jian Luan.

Technical report for the CAR-bench Challenge Open Track at IJCAI-ECAI 2026. Developed a model-agnostic framework that evolves a modular Skill Bank through trajectory-contrastive analysis and performs state-conditioned skill orchestration. The Darwin Agent team won both the **Rank Award** and **Innovation Award**, ranking **1st in the Open Track** with 70.0% Pass³ on the official hidden set.

Research Internship

Huawei Noah's Ark Lab, Research Intern

07/2025–01/2026

Worked on medical RAG and agentic reasoning: built a medical knowledge retrieval service; designed conflict-guided adaptive retrieval; fine-tuned a BERT-based answer verifier and reranker; explored reflection-based adaptive stopping and Agentic RL for medical reasoning.

Honors and Certifications

- People's Scholarship, Nanjing University (2 times).
- First-Class Academic Scholarship for master's students (2 times).
- High-value scholarship (1 time).

Academic Service

Gold Reviewer, ICML 2026; Reviewer, NeurIPS 2026.